

A Minimal LaTeX Document

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June 23, 2026

Abstract

This is a minimal LaTeX document built live during the demo. We will add sections, math, a figure, and a table step by step.

1 Introduction

LaTeX is a document preparation system that separates **content** from **formatting**. You write plain text with markup commands, and LaTeX typesets a professional PDF.

2 Methods

2.1 Statistical Model

Let Y_i be the outcome for subject $i = 1, \dots, n$. We assume the linear model:

$$Y_i = \beta_0 + \beta_1 X_i + \varepsilon_i, \quad \varepsilon_i \sim N(0, \sigma^2) \quad (1)$$

The ordinary least squares estimator is:

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y} \quad (2)$$

As shown in Equation ??, $\hat{\boldsymbol{\beta}}$ is a linear function of $\mathbf{y}$.

2.2 Lists

The data preprocessing steps are:

1. Remove duplicate records
2. Impute missing values by median
3. Standardize continuous predictors

3 Results

3.1 Descriptive Statistics

Table ?? shows the baseline characteristics of the study population.

Table 1: Baseline characteristics of study participants.

Variable	Mean	SD
Age (years)	45.2	12.1
BMI	27.3	4.8
Systolic BP	128.4	14.2
$n = 250$ participants		

4 Discussion

We have demonstrated how to write a minimal but complete scientific document in $\LaTeX$.

5 Conclusion

$\LaTeX$ provides precise, reproducible typesetting for scientific manuscripts.